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Team Name:

Bin There, Done That

Project Name:

Smart parcel sorting system



SMART PARCEL SORTING SYSTEM

Date: May 22, 2025

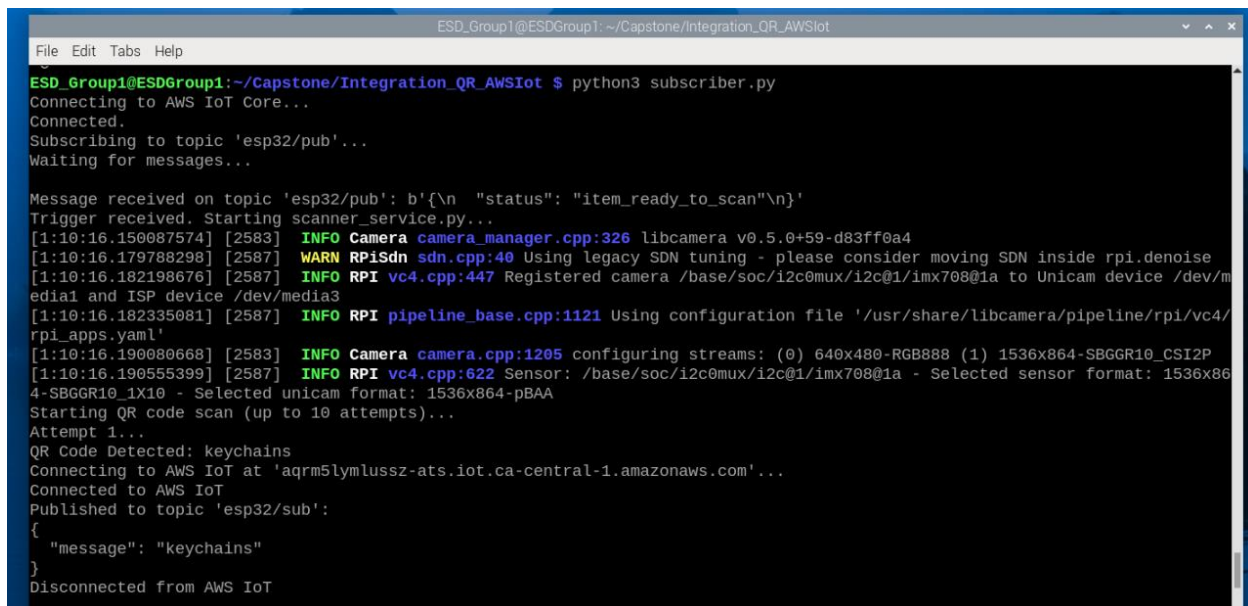
LOGBOOK WEEK 5

1. Goal Past Week

- Receive all hardware and start testing + assembling all components
- Write code to STM32 to control the conveyor and actuator (if time permit)

2. Achievement

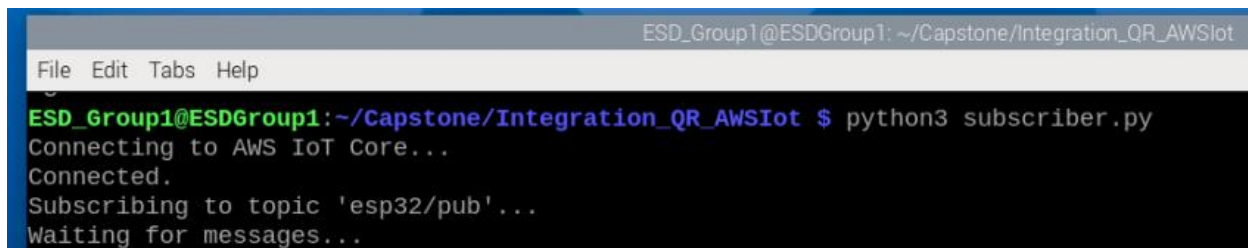
- Receive the L298N motor driver
- Successfully write and test wiring and coding to control the DC motor in the conveyor.
- Expand the Raspberry pi 4B to work as a state machine.
 - “Idle” state (waiting for esp32_pub signal)
 - “QR” state (Read QR data)
 - “Send data” state (Publish QR data to AWSIoT cloud).
 - Back to “Idle” state after publishing data.



```
ESD_Group1@ESDGroup1: ~/Capstone/integration_QR_AWSIoT
File Edit Tabs Help
ESD_Group1@ESDGroup1:~/Capstone/Integration_QR_AWSIoT $ python3 subscriber.py
Connecting to AWS IoT Core...
Connected.
Subscribing to topic 'esp32/pub'...
Waiting for messages...

Message received on topic 'esp32/pub': b'{\n  "status": "item_ready_to_scan"\n}'
Trigger received. Starting scanner_service.py...
[1:10:16.150087574] [2583] INFO Camera camera_manager.cpp:326 libcamera v0.5.0+59-d83ff0a4
[1:10:16.179788298] [2587] WARN RPiSdn sdn.cpp:40 Using legacy SDN tuning - please consider moving SDN inside rpi.denoise
[1:10:16.182198676] [2587] INFO RPI vc4.cpp:447 Registered camera /base/soc/i2c0mux/i2c@1/imx708@1a to Unicam device /dev/m
edia1 and ISP device /dev/media3
[1:10:16.182335081] [2587] INFO RPI pipeline_base.cpp:1121 Using configuration file '/usr/share/libcamera/pipeline/rpi/vc4/
rpi_apps.yaml'
[1:10:16.190080668] [2583] INFO Camera camera.cpp:1205 configuring streams: (0) 640x480-RGB888 (1) 1536x864-SBGGR10_CSI2P
[1:10:16.190555399] [2587] INFO RPI vc4.cpp:622 Sensor: /base/soc/i2c0mux/i2c@1/imx708@1a - Selected sensor format: 1536x86
4-SBGGR10_1X10 - Selected unicam format: 1536x864-pBAA
Starting QR code scan (up to 10 attempts)...
Attempt 1...
QR Code Detected: keychains
Connecting to AWS IoT at 'aqrm5lymlussz-ats.iot.ca-central-1.amazonaws.com'...
Connected to AWS IoT
Published to topic 'esp32/sub':
{
  "message": "keychains"
}
Disconnected from AWS IoT
```

Figure 1: Output of subscriber.py



```
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File Edit Tabs Help
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Connecting to AWS IoT Core...
Connected.
Subscribing to topic 'esp32/pub'...
Waiting for messages...
```

Figure 2: “Idle” state

```

QR Code Detected: keychains
Connecting to AWS IoT at 'aqrm5lymlussz-ats.iot.ca-central-1.amazonaws.com'...
Connected to AWS IoT
Published to topic 'esp32/sub':
{
  "message": "keychains"
}
Disconnected from AWS IoT

```

Figure 3: “QR” state

```

Message received on topic 'esp32/pub': b'\n "status": "item_ready_to_scan"\n}'
Trigger received. Starting scanner_service.py...
[1:10:16.150087574] [2583] INFO Camera camera_manager.cpp:326 libcamera v0.5.0+59-d83ff0a4
[1:10:16.179788298] [2587] WARN RPiSdn sdn.cpp:40 Using legacy SDN tuning - please consider moving SDN inside rpi.denoise
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4-SBGGR10_1X10 - Selected unicam format: 1536x864-pBAA
Starting QR code scan (up to 10 attempts)...
Attempt 1...
QR Code Detected: keychains

```

Figure 4: “Send data” state

2.1. Problems occur:

The camera is not fixed, and we still need to manually hold the camera

2.2. Resolve:

Buying all the mounting stuff next week to hold the camera and all the sensors plus actuator.

3. Goal for Week 6

- Receive all the hardware and start testing all components.
- Mount all the components camera, raspberry pi 4b, ...
- If time permits, start coding the state machine for STM32 to control.